

E³Sense: Head-Confined Multimodal Sensing of Learner Engagement

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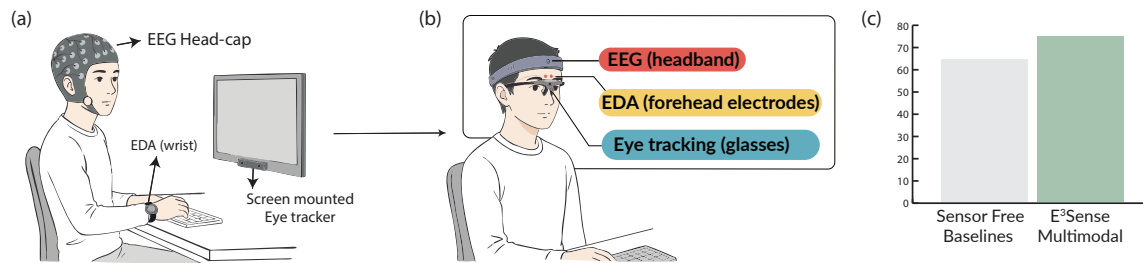


Fig. 1. **E³Sense confines engagement sensing to the head.** (a) Conventional multimodal sensing spreads electroencephalography (EEG), wrist electrodermal activity (EDA), and a screen-mounted eye tracker across body and desk. (b) E³Sense places all three on the head: an EEG headband, forehead EDA electrodes, and eye-tracking glasses. (c) Across participant-grouped folds evaluating fifteen forehead-sensing learners, E³Sense achieves a within-one-level prediction score of 75.0%, compared with 63.0% for a sensor-free reference that always predicts the most common training-fold rating. The score gives equal weight to each represented rating level. A separate exploratory analysis examines whether learner-definition codes improve prediction relative to matched score calibration (Section 7.2).

Engagement-aware learning systems could provide hints or adjust pacing when learners struggle. Prior engagement sensing work distributes sensors across or outside the body rather than consolidating them at one site, or reduces engagement to shared affect or a single dimension (from behavioral, emotional, and cognitive engagement). We introduce E³Sense, a head-worn platform that co-locates electroencephalography, eye tracking, and electrodermal activity to personalize engagement measurement. During a lab study we collected 450 ratings of engagement levels on a five-level ordinal scale while participants watched educational videos. For fifteen held-out participants, E³Sense achieved a within-one-level prediction score of 75.0%, compared with 63.0% for always predicting the most common rating. In an exploratory analysis of 18 participants from the same study who defined engagement, conditioning on learners' definitions raised the same measure by 6.9 points, from 64.6% to 71.5%. Our work provides a proof-of-concept of a head-site, personalized multimodal sensing of engagement for adaptive educational interfaces.

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CCS Concepts: • **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; *User studies*; • **Applied computing** → Interactive learning environments; • **Computing methodologies** → Cross-validation.

Additional Key Words and Phrases: engagement sensing, multimodal sensing, wearable sensing, EEG, eye tracking, electrodermal activity, learning technology

1 Introduction

Videos are widely used in online learning [30]. When learners watch prerecorded educational videos independently, an instructor may not be available to recognize confusion and adjust the pace. Engagement is associated with academic achievement [45], but different measures provide different views of the learning experience. Quiz responses, clickstream traces, pauses, and replays capture aspects of learning behavior [30], while questionnaires capture learners’ reported experiences [24, 32]. Retrospective reports depend on recall, and prompt pop-ups during viewing can interrupt the experience being measured [26, 44]. Physiological and behavioral sensing offers an additional source of continuously recorded signals. Such signals could inform systems that offer hints, adjust pacing, or suggest breaks, provided that the estimates are meaningful and available at an appropriate timescale [14, 29, 51].

However, continuous multimodal sensing can introduce substantial instrumentation and setup requirements. Prior multimodal sensing configurations distribute their sensors across the learner and the environment: a multi-electrode EEG cap on the scalp, an electrodermal sensor on the wrist or fingers, and an eye tracker mounted beneath the display (Figure 1a) [21, 34]. While such an arrangement suits laboratory experiments, maintaining and calibrating distributed sensors presents a barrier for deploying such systems in practice (e.g., watching a lecture recording in a dormitory, or attending a university course). Confining the sensor set to familiar head-worn form factors could reduce this deployment burden. This raises a feasibility question: can signals collected entirely at the head predict self-reported engagement in previously unseen learners? Implementing it with commercially available devices would also make the approach more accessible for replication and further development. We therefore investigate the technical feasibility of head-confined multimodal engagement sensing using off-the-shelf devices, providing a foundation for subsequent studies of usability and everyday deployment.

Complementing this investigation of sensor placement, we examine whether learners’ own definitions of engagement can help contextualize sensor-based predictions. Educational psychology characterizes engagement through behavioral, emotional, and cognitive dimensions [24, 26]. When providing an overall engagement rating, learners may emphasize different aspects of that experience, such as sustained attention, understanding, or emotional involvement. We explore whether these self-described criteria provide useful context for mapping physiological and behavioral signals to learners’ engagement ratings.

In this paper we present **E³Sense** (Engagement via EEG, eye tracking, and EDA), a head-worn multimodal sensing platform. E³Sense co-locates three devices on the head: a dry-electrode EEG headband, eye-tracking glasses, and an EDA sensor with dry forehead electrodes. It aligns their three streams in software and predicts a learner’s segment-level engagement rating from a single fused representation of all three (Section 3). All sensing sites lie on the head, and the EEG and EDA electrodes are dry. This layout could inform a future integrated device. To evaluate E³Sense, we ask two research questions.

- **RQ1:** How accurately does E³Sense predict engagement for unseen learners, and how does performance vary across conventional model families?

- **RQ2:** How do learners define engagement, and what is observed when those definitions are supplied as context to an exploratory calibration model?

Two analyses of the same study answer these questions under separate evaluation protocols (Section 6). RQ1 is the paper’s central protocol: five participant-grouped folds train across both EDA collection sites and score the fifteen held-out forehead participants so every rating scored comes from a learner no model was fitted to. RQ2 is a smaller, exploratory evaluation over the eighteen participants who gave definitions of engagement in their own words, under its own six-fold protocol. The two studies draw on different cohorts and different procedures, so we report them separately and do not compare their numbers with each other.

E³Sense achieves 75.0% balanced one-off accuracy, which gives equal weight to each engagement class and counts predictions within one level of the learner’s rating as correct. This exceeds the sensor-free baseline of 63.0% by 12.0 percentage points (Section 7.1). In the complementary analysis, learners described engagement through cognitive, behavioral, and emotional criteria. Incorporating these criteria into an exploratory calibration model increased balanced one-off accuracy from 64.6% to 71.5% relative to a matched model without this context (Section 7.2). Together, these findings support the technical feasibility of head-confined multimodal engagement sensing and motivate further investigation of learner definitions as context for personalized engagement prediction.

Our contributions are:

- (1) **E³Sense, a head-worn multimodal sensing platform, and a participant-independent evaluation of head-confined engagement prediction.** E³Sense co-locates dry-electrode EEG, eye tracking, and dry forehead EDA on the head and fuses them into a fixed 166-dimensional per-segment representation. We evaluate that representation with five classical model families and sensor-free baselines on held-out forehead participants. AdaBoost attains $75.0\% \pm 7.0$ balanced 1-off accuracy and 1.043 ± 0.158 macro-MAE, a fold mean 12.0 percentage points above the mode baseline, and LightGBM holds the highest binary macro-F1 point estimate at $58.9\% \pm 7.9$ (§7.1).
- (2) **Learner definitions of engagement, coded qualitatively, with an exploratory predictive probe.** Participants’ coded accounts span cognitive, behavioral, and emotional definitions. Supplying those codes as context to the second-stage ordinal model attains 71.5% balanced 1-off accuracy against 64.6% for a matched control, an observed +6.9-percentage-point difference alongside lower macro-MAE. Within this eighteen-participant cohort, what a learner counts as engagement carries information about their ratings that the sensor-derived scores alone do not (§7.2).

2 Background and Related Work

Prior work relevant to E³Sense spans three linked questions: how engagement is conceptualized, how engagement labels are elicited and evaluated, and how physiological sensing modalities are configured. We review these threads to motivate two gaps: (i) whether multimodal engagement sensing can be confined to the head while retaining participant-independent predictive utility, and (ii) whether learner-specific definitions can provide context for the target being predicted.

2.1 Background on Engagement

Engagement is a complex multi-dimensional construct, where its specific components and associated metrics are domain-dependent (see Doherty et al. for a review of engagement theories and measurement [20]). One framework commonly used in educational psychology is the *tripartite framework* [23], which identifies three interconnected aspects

of engagement: behavioral participation and attention, emotional interest and enjoyment, and cognitive effort and depth of processing [23]. These dimensions are related but dissociable [5, 7] and have associations with academic achievement [45]. We use this framework to motivate E³Sense and to discuss learners’ accounts, while targeting a single overall self-reported engagement rating as our prediction outcome. Automatic engagement detection has also been studied through face- and gaze-based recognition during learning [17, 36, 59], affect-in-the-wild competitions [18], and reviews of multimodal learning-experience data [29] and wearable sensing for engagement specifically [14]. We build on this literature’s physiological and behavioral signals.

2.2 Engagement Labels and Evaluation

Engagement is commonly measured through questionnaires, post-task surveys, and experience sampling [24], including standardized affect instruments such as the Self-Assessment Manikin [9]. The temporal unit matters. A session-level response summarizes more activity than a segment rating, while retrospective reports cannot directly resolve moment-to-moment changes [26]. Physiological and behavioral signals offer continuous correlates, but relating them to a rating depends on what the rating denotes [27, 28]. We collect learners’ own accounts to contextualize that target and examine their predictive use.

Engagement labels are often unevenly distributed [31, 56]. Frequency-weighted performance and class-balanced ordinal performance answer different questions [2]. We report balanced 1-off accuracy and macro-MAE alongside sensor-free baselines, with binary macro-F1 as a secondary sensitivity analysis (Section 6.1). This makes the contribution of the label distribution explicit without treating near-correct ordinal predictions as exact five-class decisions.

Participant-disjoint evaluation is also consequential. Physiological variation between people can make performance on familiar participants different from performance on unseen ones [55]. Engagement studies have used both participant-disjoint protocols [19, 21, 28, 49] and segment-level splits [61]. We hold participants out whole in our outer evaluation, the stricter protocol and the one that speaks to how the model generalizes to new learners. RQ2 uses their exit-interview definition codes as context.

2.3 Multimodal and Head-Worn Sensing

EEG, eye tracking, and EDA provide candidate correlates of engagement. EEG band power is associated with attention and cognitive processing [40, 51], including in classroom settings [53]; gaze describes visual orientation and pupil dynamics reflect several influences, including cognitive load [4, 35, 54]. EDA reflects sympathetic activation [8]. Multimodal physiological datasets and frameworks combine EEG with other channels for affect recognition [42, 62], but none provides a unique reading of one engagement dimension. We therefore evaluate the complete fused E³Sense representation against the same overall rating across five classical model families.

Placement distinguishes existing configurations. SenseSeek combines head-worn EEG, wrist EDA, and screen-mounted eye tracking for information seeking [34]. Effecti-Net combines head-worn EEG and eye tracking with finger EDA and PPG to predict perceived content effectiveness [21]. Long et al. combine EEG and eye tracking to detect internal versus external attention in VR [46], and Brishtel et al. combine eye tracking and EDA to detect mind wandering during reading [13]. Head-worn platforms such as Galea and HMDBioPad demonstrate co-located instrumentation [6, 57], while scalp EEG with forehead EOG has been evaluated for vigilance [63]. These motivate examining what EEG, eye tracking, and forehead EDA together predict about engagement during learning.

Table 1 summarizes placement, targets, and labeling practices in the studies considered here. Of these, Knierim et al. [41] and Kosmyrna et al. [43] confine every sensing site to the head but each uses a single sensing modality (EEG),

while Galea [6] and HMDBioPad [57] are head-confined multimodal platforms not evaluated for engagement prediction. E³Sense is, to our knowledge, the first configuration in this comparison set to combine three head-confined modalities, evaluate them against segment-level engagement ratings, and pair the sensing results with learners’ own definitions of engagement. We compare five classical model families on the same fused, handcrafted-feature representation, keeping the sensor input fixed throughout RQ1.

Table 1. Comparison of Student Engagement Sensing and Labeling Practices in Prior Work.

Work	Data Sources	Head-Confined	Eng. Framing	Duration	Feedback Type	Temporal Resolution	Eng. Granularity
Di Lascio et al. [19]	EDA (wrist)	No	Emotional	45 min	Std. Q.	Lecture	Binary
Monkarese et al. [49]	Video	No	Behavioral	60 min	Self-report	2 min / 10 s	Binary
Dwivedi et al. [21]	EDA/PPG-f; eye/EEG-h; cam	No	—	9 min	Ratings	Segment	Likert-3
Knierim et al. [41]	EEG (head)	Yes	Cognitive	45 min	Std. Q.	Task	Ternary
Ji et al. [34]	EDA-w; eye (screen); EEG-h	No	—	—	—	—	—
Kosmyna et al. [43]	EEG-h; haptic scarf (neck)	Yes[†]	Cognitive	105 min	Self-report + quiz	Slide	Binary
Gao et al. [28]	EDA/HRV-w; env. sensors	No	Behav./ emot./cog.	40–80 min	Std. Q.	Class	Likert-5
Xiao et al. [61]	EEG-h; eye (screen); cam	No	Behav./ emot./cog.	40 min	Std. Q.	Session	Likert-5
Bernal et al. [6]	Physiological suite (head, HMD)	Yes	None	—	—	—	—
Wan et al. [57]	Bio-signal pads (head, HMD)	Yes	None	—	—	—	—
E³Sense (this paper)	EEG/eye/EDA (head)	Yes	Overall rating; three-dimension definitions	36 min	Self-report + interview	Segment	Likert-5

Abbreviations: w = wrist; f = finger; h = head; cam = webcam; env. = environmental; cog. = cognitive; behav. = behavioral; emot. = emotional; std. Q. = standardized questionnaire. The temporal-resolution column reports the unit each engagement label covers. [†]AttentivU’s scarf delivers haptic feedback and carries no sensor, so every sensing site is on the head. Dashes mark properties the work does not report; where a system targets a construct other than engagement, as SenseSeek and Effecti-Net do, the engagement-specific columns do not apply to it.

3 The E³Sense Sensing Configuration

3.1 Design Goals

The configuration was designed to co-locate neural, ocular, and electrodermal sensing on the head and evaluate the complete configuration in a controlled video-learning study. Dry electrodes and head-worn devices make this placement practical to deploy.

3.2 Architecture

E³Sense co-locates three devices on the head and aligns their streams in software, establishing a sensor layout that a future integrated device can adopt directly. Figure 1 diagrams the sensor placement; Section 5 covers feature extraction and prediction.

3.2.1 Sensing Layer. Three consumer-grade devices supply the three streams.

- **EEG headband.** The Frenz BrainBand [22] delivers four dry-electrode channels over Bluetooth at 125 Hz: two frontal (LF, RF) and two temporal over the ears. Only the frontal pair enters the analysis, for its established sensitivity to attention regulation and cognitive control [51]. We use dry electrodes for faster, gel-free setup [15]; band-power features of the kind we extract remain reliable with dry electrodes [39].
- **Eye tracker.** The Pupil Labs Neon [52] captures infrared eye video at 200 Hz and processes it on-device into a twenty-channel 3D eye state: pupil diameter, eyeball-center and optical-axis coordinates for both eyes, and eyelid angles and apertures. We retain all twenty.
- **EDA sensor.** The EmotiBit [50] measures EDA at 15 Hz. To move the measurement to the head we replaced its electrodes with two dry silver/silver-chloride electrodes on the forehead just above the eyebrows, the eye-tracker frame fixing their separation at ≈ 4 cm so spacing is identical across participants. Prior comparisons characterize forehead EDA as relatively resistant to motion artifacts but less responsive than palmar recordings, motivating a feasibility evaluation rather than an equivalence claim [33].

The three devices clock independently, so we align the streams on device-native timestamps and periodic clock anchors collected during recording, making features from one window traceable to the same stretch of viewing in all three.

3.2.2 Preprocessing Layer. Each stream reaches the feature extractor filtered for its sensor’s characteristic artifacts, with the frequency content its features depend on retained. For EEG, the headband delivers the frontal channels band-limited from 0.5 to 60 Hz, a passband chosen on standard filter-design grounds [60]: it removes the DC drift that dry contacts introduce while retaining the canonical bands (δ 1 to 4 Hz, θ 4 to 8, α 8 to 13, β 13 to 30, and γ 30 to 45 Hz), from whose token spectrum we compute band power. For eye tracking, we use the twenty eye-state channels directly as the device exports them [52]. The eye features in Table 2 are therefore statistics of these per-sample channels plus a few derived pupil and blink quantities; where a threshold is needed, we take it from the token’s own median. For EDA, a fourth-order Butterworth lowpass at 0.5 Hz attenuates higher-frequency components while retaining the tonic conductance level and the slow phasic envelope [8]; we decompose the filtered signal into these tonic and phasic components, both of which carry information about sympathetic arousal [47].

3.2.3 Tokenization. Two units of time recur here and are distinct. A *segment* is the rated unit of Section 4. An *analysis window* is the fixed 120 s interval from the start of each segment, from which we compute that segment’s features, so every observation carries an equal span of signal regardless of video length. Within that window we use twelve 10 s EEG tokens, six 20 s eye-tracking tokens, and three 40 s EDA tokens. EEG band power uses Welch estimates over 4 s sub-segments. Eye tokens summarize the device-exported channels and derived pupil and blink quantities. EDA tokens use a 10 s centered rolling median for the tonic component. Features are mean-pooled across tokens within each modality, giving one vector per segment’s analysis window.

4 Study Overview

The study addressed two limitations of common engagement-measurement approaches. First, session or video level ratings can obscure changes in engagement within a video. We therefore asked participants to retrospectively rate five segments after each video. Second, standardized questionnaires define engagement in advance and may not capture how individual learners interpret the construct. We therefore conducted semi-structured exit interviews in which participants described what engagement meant to them and how they formed their ratings. Both components were completed within

a single study session (Figure 2). A single session served both questions. We first describe the collection procedure common to them, then the sensor placement that RQ1 turns to, and finally the interviews that RQ2 draws on.

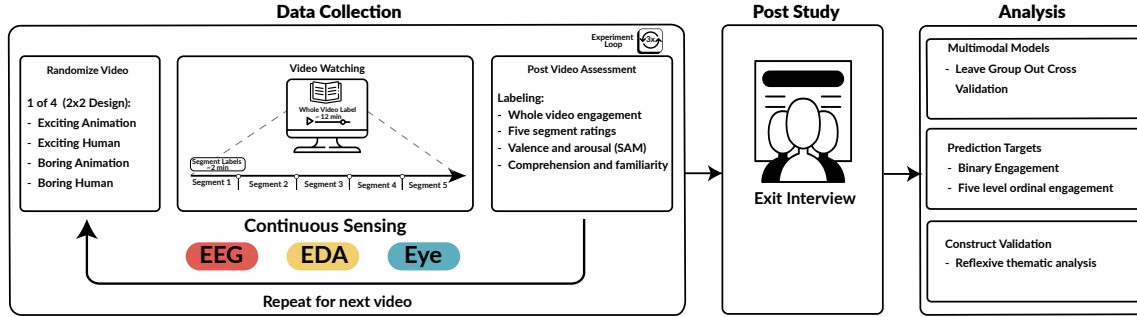


Fig. 2. Collection and analysis procedure. Participants watched three of four videos in randomized order under continuous EEG, eye, and EDA sensing, completing a post-video assessment that includes five segment ratings and additional video-level measures; the loop repeats three times. The leading 120 s of each rated segment supplies the modeled features. EDA was recorded at the wrist for the first fifteen participants and the forehead for the next fifteen. 24 participants additionally completed an exit interview out of which 18 gave definitions of engagement in their own words. Five classical model families using the fused E³Sense representation are then evaluated under participant-grouped cross-validation, predicting binary and five-level ordinal engagement.

4.1 Data Collection Common to Both Questions

4.1.1 Participants and Set-up. We recruited thirty participants ($N = 30$) from a university campus. Ages ranged from 19 to 43 years (median 26, $SD = 5.86$); 11 were female and 19 male. We required English fluency, vision normal or corrected by contact lenses, and no history of neurological conditions. Eyeglasses occlude the eye tracker’s infrared cameras, and neurological history can introduce confounds in EEG signals. Each received a \$20 gift card for about 60 minutes. The institution’s Institutional Review Board (IRB) approved the protocol. On arrival participants watched a device-fitting demonstration, gave informed consent, then sat at a 27-inch monitor and put on the recording devices, whose signal quality we verified before viewing began.

4.1.2 Video Stimuli and Assignment. Four publicly available educational videos populate a 2×2 design crossing content type (Exciting/Boring) with presentation format (Animation/Human), one per cell: Exciting Animation (12:08, 728 s), Exciting Human (12:54, 774 s), Boring Animation (12:04, 724 s), and Boring Human (11:41, 701 s), covering human anatomy, gene modification, popular science, and linear algebra. Content type was assigned a priori, before any data collection: four researchers viewed the candidates and settled by discussion which were exciting and which boring.

The four videos vary content and presentation. The four cells were used to diversify the viewing stimuli rather than to support factorial inference; with one video per cell, we do not estimate independent effects of content type or presentation format. The roughly 12-minute length sits well above the under-six-minute chunks MOOC engagement research recommends [30] – one reason we rate per segment. Each participant watched three of the four, selection and order randomized in Qualtrics, giving 90 video observations: Exciting Animation ($N = 21$), Exciting Human ($N = 25$), Boring Animation ($N = 21$), and Boring Human ($N = 23$).

4.1.3 Engagement Labeling. Participants rated engagement retrospectively, immediately after each video rather than continuously during viewing, following standard practice [24]; this keeps the viewing itself uninterrupted and more

natural, without introducing confounding factors that would affect the state we want to capture. They first gave an overall engagement rating on a five-point Likert scale (Extremely Disengaged to Extremely Engaged). This granularity was chosen to distinguish intensities without imposing too much load across repeated ratings [44]. Then they rated engagement for five equal segments: for a video of duration T , segment k spans $[kT/5, (k+1)T/5]$ for $k = 0, \dots, 4$, some 140 to 155 s. The cohort supplies 450 segment-level observations, the prediction target throughout, because the whole-video rating averages over exactly the fluctuations an engagement sensing system wants to detect.

We code the five categories as 0 (Extremely Disengaged), 1 (Disengaged), 2 (Neutral), 3 (Engaged), and 4 (Extremely Engaged). The label distribution skews toward engagement: 37, 57, 73, 169, and 114 observations from *Extremely Disengaged* to *Extremely Engaged*, so 63% of the ratings sit at the two engaged levels and 21% at the two disengaged ones. We therefore report the main prediction metrics in Section 6.1 with class-balanced averaging, so performance on the rarer disengaged states is not masked by the majority classes.

4.2 RQ1: Head-Confined E³Sense Evaluation

Confining the sensor set to the head moves the EDA electrodes from the wrist to the forehead (Section 3). Fifteen participants wore wrist EDA, the conventional site [19, 28], and the remaining fifteen wore forehead EDA. No participant wore both, so we report no wrist–forehead contrast. Models train on both sites because head confinement constrains where sensors sit during use, not where the training data came from. Every learned RQ1 row uses the same 166-dimensional E³Sense representation. Every model-family result in Table 3 is scored on the fifteen forehead participants alone (Section 6.2.1). The definition-context analysis scores all eighteen coded participants, the seven wrist-EDA participants included, since its cohort follows the interviews, not sensor site (Section 6.3.2).

4.3 RQ2: Collecting Learner Definitions

Six participants out of 30 left the session before the interview could be finalized, due to unforeseeable conflicts or time constraints. The remaining participants completed semi-structured interviews and a subset of 18 participants defined engagement in their own words. In the interviews we asked how they had decided their ratings, what raised or suppressed their engagement, and how they would describe engagement in their own words. These accounts supply the learner-defined descriptions of engagement that RQ2 draws on, as well as context for interpreting the quantitative ratings.

5 Algorithms

E³Sense extracts handcrafted features from each stream and concatenates them into a fixed representation for classical prediction models. Below we describe: (i) the preprocessing and feature-extraction pipeline shared by both questions, (ii) the fused E³Sense representation and five model families evaluated in RQ1, and (iii) the second stage through which a learner’s own definition of engagement enters the RQ2 model.

5.1 Common to Both Questions

5.1.1 Modeling Overview. The 120 s analysis window of Section 3.2 is set by the 40 s EDA token. The window must hold a whole number of them and still fit inside the shortest segment of 140 s, and 120 s also divides evenly into the 10 and 20 s tokens. Within each analysis window the pipeline runs in three stages (Figure 3): (i) preprocessing and tokenization (Section 3.2.2), (ii) feature extraction and mean-pooling, and (iii) task-specific prediction. Every RQ1 model

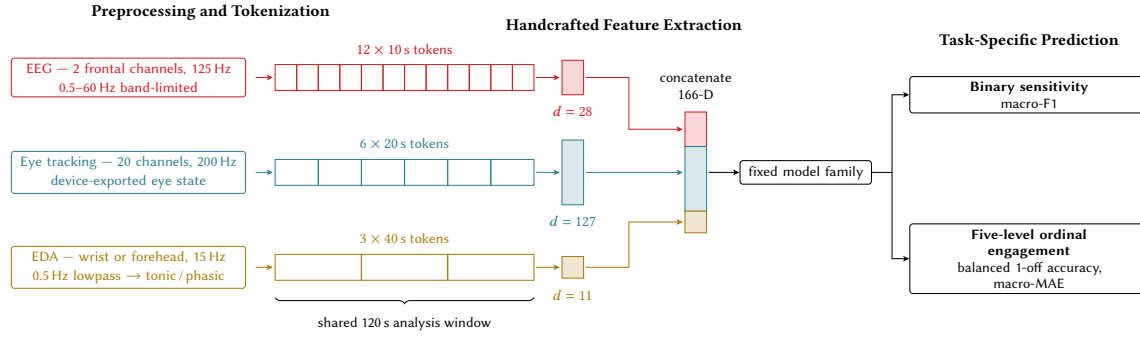


Fig. 3. The E³Sense processing pipeline. We preprocess each stream, tokenize it, compute handcrafted features from every token, mean-pool them into one vector per modality and window, and pass the result to classical models trained separately for binary and five-level ordinal engagement. RQ1 evaluates five model families as separate rows, keeping the 166-dimensional E³Sense input fixed (Section 6.2.2).

family receives the same 166-dimensional fused representation and is evaluated under the same protocol. RQ2 adds a second stage over the pipeline’s outputs rather than altering the pipeline itself (Section 5.3).

5.1.2 Handcrafted Feature Extraction. We extracted the handcrafted features summarized in Table 2 independently from each sensor-specific token and mean-pooled corresponding features across the 120 s analysis window. The 28-dimensional EEG representation combines time-domain statistics, Hjorth parameters, frequency-band power estimated using Welch’s method, and spectral entropy [46, 58]. The 127-dimensional eye-tracking representation summarizes the twenty device-exported eye-state channels together with derived blink, gaze-dispersion, and pupil-dynamics measures used in related attention-sensing work [13, 46]. The 11-dimensional EDA representation characterizes the raw, tonic, and phasic signals using statistical, response-count, amplitude, area, and trend features previously applied in classroom and engagement sensing [28]. Concatenating the three representations yields the fixed 166-dimensional E³Sense vector supplied to every learned RQ1 model.

5.1.3 Data Quality Check. Following established EDA quality-assessment practices [1, 38], we excluded four participant-session EDA recordings—two wrist and two forehead—because the recording was missing owing to battery failure or contained fewer than 30 finite samples, our prespecified minimum-data threshold. The four sessions account for 20 of the 450 study segments. Two affected sessions belong to forehead participants, producing 10 rows among the 225 RQ1 test segments (4.4%) without measured EDA. All eleven EDA features are missing in affected rows, so we impute each feature from its median among the training participants in that fold. EEG and eye-tracking features are complete for every segment. Section 7.1 reports a sensitivity analysis that excludes affected sessions from model fitting and evaluation.

5.2 RQ1: Models for the Head-Confined E³Sense Representation

5.2.1 E³Sense Feature Representation. For multimodal prediction we use early feature-level fusion [3], concatenating the mean-pooled EEG, eye-tracking, and EDA feature vectors for the same analysis window:

$$\mathbf{x}_{\text{multi}} = [\mathbf{x}_{\text{EEG}}; \mathbf{x}_{\text{Eye}}; \mathbf{x}_{\text{EDA}}], \quad (1)$$

Table 2. Handcrafted features extracted from each modality. We compute features within every temporal token and mean-pool them across the analysis window, giving one vector per modality and window; d is the resulting dimensionality.

Modality	d	Features
EEG	28	Per frontal channel (LF, RF), 14 each: mean, standard deviation, root-mean-square and peak-to-peak amplitude, zero-crossing rate, Hjorth activity, mobility and complexity, log band power in the canonical bands (δ 1 to 4 Hz, θ 4 to 8 Hz, α 8 to 13 Hz, β 13 to 30 Hz, γ 30 to 45 Hz), and spectral entropy of the normalized power spectrum.
Eye tracking	127	Six statistics (mean, standard deviation, peak-to-peak range, mean absolute successive difference, and the 25th and 75th percentiles) for each of twenty per-sample channels: pupil diameter, eyeball-center and optical-axis coordinates (x , y , z), eyelid angles (top and bottom) and eyelid aperture, left and right (120 dimensions). Derived, left eye: blink fraction, the share of samples whose eyelid aperture falls below half its token median; gaze dispersion, the summed standard deviation of the optical-axis x and y components; mean pupil dilation and constriction rate; the fraction of invalid pupil samples; and the mean and standard deviation of pupil diameter (7 dimensions).
EDA	11	Mean and standard deviation of the raw signal and of its tonic and phasic components; skin conductance response (SCR) count; mean and maximum SCR amplitude; area under the positive phasic curve; and the linear trend across the window.

giving a 166-dimensional representation from 28 EEG, 127 eye-tracking, and 11 EDA features. This complete vector is the fixed input for every learned RQ1 row.

5.2.2 Model Families. We evaluate a linear model, LightGBM [37], XGBoost [16], AdaBoost [25], and Random Forest [12]. These families are commonly used with handcrafted physiological features (Section 2.3), and the four tree ensembles suit a few hundred labeled observations with mixed scales and correlated predictors. Each family appears in a classifier or regressor variant according to the target, with fixed within-family hyperparameters. Each family is reported separately and remains fixed across the five outer folds.

5.2.3 Sensor-Free Baselines. We additionally evaluate three sensor-free baselines using only the labels available in each training fold:

- *Random distribution:* We draw predictions from the training-fold label distribution and average the metric over 400 draws within each evaluation fold, with the fold index as the random seed.
- *Rounded mean:* We assign every test instance the rounded mean of the training-fold labels for the task.
- *Mode:* We assign every test instance the most frequent label in the training fold.

We recompute each baseline within every fold, and none uses a physiological signal. They therefore provide label-only reference points for interpreting model performance. A sensor-based model should be compared against these baselines to determine whether its predictions add information beyond the fold’s label distribution. We report all three baselines, combining the mode and rounded-mean rows when their saved predictions coincide (§8.2.3 discusses mode’s aggregation properties).

5.3 RQ2: Definition Codes as Model Input

RQ2’s second half asks whether supplying a learner’s own definition of engagement improves engagement detection. We enter that definition as an attribute of the participant rather than as sensor input: the interview text itself never

reaches the model, only the Cognitive, Behavioral, and Emotional codes recovered from it (Section 6.3.1), coded as three binary indicators per participant.

The codes enter only a second-stage ordinal calibrator — a five-level cumulative-link model [48] that models the levels explicitly — fit over four per-modality engagement scores (fused-modality, EEG, eye, EDA) plus the site indicator (Figure 4). All three calibration arms are fit on identical upstream inputs, the same frozen, cross-fitted training scores and out-of-sample test scores within each outer partition (Section 6.3.2), so any difference between arms comes only from the code covariates each one adds. These four scores are internal RQ2 covariates and are not interpreted as RQ1 sensor-ablation results.

The gating arm additionally includes an interaction between each of the three definition codes and each of the four sensor-derived scores, resulting in 12 interaction terms. A matched control — the four scores and the site indicator, with no codes — supplies the reference against which both definition arms are read. Two further baselines complete the comparison. The fused-modality score used on its own, with no calibrator inputs, and a constant predictor fixed at ordinal level 3.

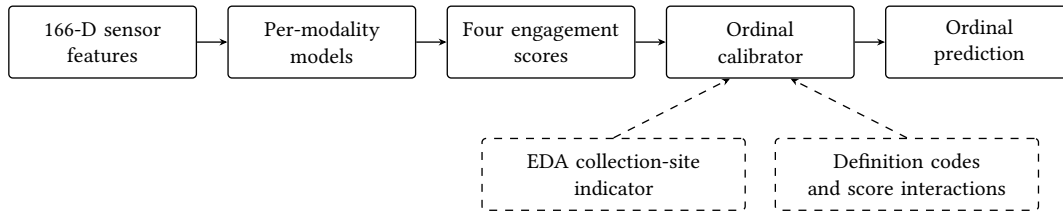


Fig. 4. Definition context as model input. Within each of the six folds, sensor features pass through that fold’s per-modality models, yielding four engagement scores. A held-out participant’s scores are that fold’s out-of-sample predictions for them. A second-stage ordinal calibrator receives those scores and the EDA collection-site indicator. The definition-conditioned arm additionally receives the cognitive, behavioral, and emotional codes and their interactions with the four scores. The codes do not enter the 166-D feature vector. Section 6.3.2 describes how upstream scores are generated and used during calibrator validation.

6 Evaluation Methodology

We evaluate two claims under participant-independent protocols. RQ1 tests how accurately the complete head-confined E³Sense configuration predicts segment-level engagement and how performance varies across fixed classical model families. RQ2 tests whether learners’ own definitions of engagement improve the calibration of these predictions. In both evaluations, each participant is assigned to a single outer test fold and is excluded from all model fitting for that fold. RQ1 trains on participants from both EDA sites and evaluates held-out forehead participants. RQ2 uses the eighteen participants who provided explicit engagement definitions and constructs its upstream predictions within each RQ2 training partition.

6.1 Targets and Metrics

The primary target is the five-level segment rating, coded 0–4. We also conduct a binary sensitivity analysis in which codes 3 (Engaged) and 4 (Extremely Engaged) are positive and codes 0–2 are negative. Binary performance is measured by macro-F1, the mean F1 across the two classes.

For ordinal prediction, let $C_{\text{test}} = \{c : N_c > 0\}$ denote the classes represented in a scored fold and let N_c denote the number of samples in class c . We report balanced 1-off accuracy and macro-MAE:

$$\text{B1Off} = \frac{1}{|C_{\text{test}}|} \sum_{c \in C_{\text{test}}} \frac{1}{N_c} \sum_{i: y_i = c} \mathbb{I}(|\hat{y}_i - y_i| \leq 1), \quad (2)$$

$$\text{MacroMAE} = \frac{1}{|C_{\text{test}}|} \sum_{c \in C_{\text{test}}} \frac{1}{N_c} \sum_{i: y_i = c} |\hat{y}_i - y_i|. \quad (3)$$

Balanced 1-off accuracy gives equal weight to each represented class and credits predictions within one ordinal level of the target. Macro-MAE gives equal class weight to absolute ordinal error.

6.2 RQ1: Engagement from the Head-Confined E³Sense Configuration

6.2.1 Participant-Independent Evaluation. We use five participant-grouped outer folds. The sorted list of 30 participants is shuffled and assigned to five groups of six. Each fold trains on the remaining 24 participants from both EDA sites and evaluates the forehead participants in the held-out group.

The evaluation set contains 225 segments: 24, 31, 34, 84, and 52 for ordinal codes 0–4, respectively. The binary analysis contains 136 positive and 89 negative segments. This protocol matches the target setting of mixed-site model development followed by forehead-only deployment.

6.2.2 Model Development. For each prediction task, we independently fit and evaluate a linear model, LightGBM, XGBoost, AdaBoost, and Random Forest on the same 166-dimensional E³Sense representation. Each family is a separately reported row, and its identity and within-family hyperparameters remain fixed across the five outer folds. Every family uses the same participant assignments in each fold.

All preprocessing is learned from the relevant outer training split. This includes EDA median imputation and standardization. The linear, LightGBM, and Random Forest classifiers use balanced class weights. The other classifiers use their configured defaults. Ordinal regressors produce a continuous score that is rounded to the nearest integer and clipped to codes 0–4. We report descriptive equal-fold means and population standard deviations; RQ1 makes no null-hypothesis significance claim.

As a secondary sensitivity analysis, we repeat the fixed-family comparison with fold-local top-20 feature selection. Within each outer fold and model family, we fit an initial model on the 24 training participants, rank the 166 inputs using model-derived importance (or absolute coefficient magnitude for the linear model), retain the 20 highest-ranked features, and refit the same family on the reduced representation. EDA imputation, scaling, and feature ranking are learned from the outer-fold training data only; the outer-test participants inform none of these steps. The feature ranking is recomputed independently in every outer fold. All five families are reported separately, and the top-20 cutoff is fixed.

6.3 RQ2: Learner Definitions of Engagement

6.3.1 Thematic Analysis and Definition Codes. We analyzed the interview transcripts using reflexive thematic analysis [10, 11]. One author conducted the initial coding and developed an inductive coding frame. Two additional authors reviewed the interpretations, proposed alternatives, and resolved divergent readings through discussion. The alignment with the tripartite framework emerged from this analysis.

Eighteen of the 24 interviews contained an explicit definition of engagement. We represented these definitions with three overlapping participant-level indicators: Cognitive (seven participants), Behavioral (five), and Emotional (eight). The six interviews without an explicit definition were excluded from the prediction analysis.

6.3.2 Definition-Conditioned Prediction. RQ2 uses the eighteen coded participants and their 270 segments. Six outer folds, grouped by participant, each hold out three participants (45 segments). All eighteen participants are evaluated once.

Within each outer fold, modality-specific models generate cross-fitted scores for the fifteen training participants and out-of-sample scores for the three test participants. These models use 166 mean features and AdaBoost regressors.

The second-stage ordinal calibrator receives four scores—fused modality, EEG, eye tracking, and EDA—plus EDA site. We compare this score-calibration model with two definition-aware variants. The first adds the three definition indicators as main effects. The second adds twelve interactions between the four scores and three definition indicators, for a total of 20 design columns. The raw fused-modality score provides an additional reference.

7 Results

Table 3. **Participant-independent engagement prediction with E³Sense.** Every learned model receives the same 166-dimensional E³Sense representation. Values are equal-fold means $\pm$ population SD across five outer folds, with model family fixed across folds. Ordinal metrics are primary, and binary macro-F1 is secondary. Random-distribution results average 400 draws per fold. Higher is better for balanced 1-off accuracy and binary macro-F1; lower is better for macro-MAE. Bold marks the best point estimate in each column.

Group	Model	Balanced 1-off (%)	Macro-MAE	Binary macro-F1 (%)
Sensor-free	Mode / rounded mean	63.0 $\pm$ 6.0	1.320 $\pm$ 0.160	37.6 $\pm$ 4.4
	Random distribution	54.9 $\pm$ 3.3	1.520 $\pm$ 0.092	49.0 $\pm$ 1.1
E ³ Sense	Linear model	57.6 $\pm$ 16.8	1.239 $\pm$ 0.258	50.5 $\pm$ 11.2
	Random Forest	67.3 $\pm$ 5.1	1.161 $\pm$ 0.097	47.2 $\pm$ 4.6
	XGBoost	70.0 $\pm$ 5.9	1.106 $\pm$ 0.126	56.7 $\pm$ 5.1
	LightGBM	72.9 $\pm$ 5.5	1.078 $\pm$ 0.115	58.9 $\pm$ 7.9
	AdaBoost	75.0 $\pm$ 7.0	1.043 $\pm$ 0.158	54.3 $\pm$ 3.0

7.1 RQ1: Engagement from the Head-Confined E³Sense Configuration

7.1.1 E³Sense Model-Family Comparison. AdaBoost has the strongest ordinal point estimates in Table 3, reaching 75.0% $\pm$ 7.0 balanced 1-off accuracy and 1.043 $\pm$ 0.158 macro-MAE. Relative to the mode baseline, these equal-fold means differ by +12.0 percentage points in balanced 1-off accuracy and -0.277 in macro-MAE. These comparisons are descriptive.

The other tree ensembles attain balanced 1-off means from 67.3% to 72.9%, whereas the linear model reaches 57.6%. LightGBM has the second-highest ordinal estimates, with 72.9% $\pm$ 5.5 balanced 1-off accuracy and 1.078 $\pm$ 0.115 macro-MAE. The ordering changes for the secondary binary target: LightGBM has the highest macro-F1 point estimate at 58.9% $\pm$ 7.9, followed by XGBoost at 56.7% $\pm$ 5.1 and AdaBoost at 54.3% $\pm$ 3.0.

7.1.2 Sensitivity Analyses. The binary results show that model-family rankings depend on whether engagement is represented as a five-level ordinal outcome or collapsed into two classes. For reference, the random-distribution binary baseline reaches $49.0\% \pm 1.1$ macro-F1.

We also refit every family after excluding the four participant-session recordings without measured EDA from both training and evaluation. This leaves 215 test segments. AdaBoost remains the highest ordinal point estimate across the fixed families, reaching $72.7\% \pm 4.8$ balanced 1-off accuracy and 1.066 ± 0.131 macro-MAE. This complete-case result is a descriptive missing-data sensitivity analysis.

The fixed-family feature-selection results in Table 4 show a family-dependent pattern. Retaining the top 20 features improves both ordinal metrics for the linear and Random Forest models, but worsens both for XGBoost and LightGBM. For AdaBoost, balanced 1-off accuracy increases by 0.9 percentage points, from 75.0% to 75.9%, while macro-MAE increases by 0.005, from 1.043 to 1.048, and binary macro-F1 decreases by 1.3 percentage points, from 54.3% to 53.0%. Feature-selection effects are therefore mixed rather than uniformly beneficial, supporting the complete 166-dimensional representation as the primary analysis.

Table 4. **Fixed-family performance with fold-local top-20 feature selection.** Values are equal-fold means $\pm$ population SD across five participant-grouped outer folds. Feature rankings are learned only from each outer fold’s training participants.

Model family	Balanced 1-off (%)	Macro-MAE	Binary macro-F1 (%)
Linear model	66.2 ± 10.1	1.141 ± 0.156	53.7 ± 12.7
Random Forest	72.4 ± 9.9	1.089 ± 0.151	46.7 ± 6.0
XGBoost	65.3 ± 9.5	1.201 ± 0.126	52.6 ± 12.1
LightGBM	67.7 ± 4.4	1.161 ± 0.106	58.0 ± 6.5
AdaBoost	75.9 ± 3.6	1.048 ± 0.120	53.0 ± 3.0

Figure 5 characterizes the AdaBoost importance ranking that drives its fold-local top-20 refit.

7.2 RQ2: Learner Definitions of Engagement

7.2.1 How Participants Define Engagement. Participants defined engagement through learning and understanding, sustained attention, interest, and emotion. These themes align with the cognitive, behavioral, and emotional dimensions of the tripartite framework [23], while showing that participants relied on different criteria when rating their engagement.

Cognitive accounts emphasized effort and knowledge gained: *“I think for me it would be like whether teaching me something new...In the end, I am learning something”* (P18). Behavioral accounts emphasized sustained attention: *“As long as my thought didn’t get distracted and as long as I didn’t zone out of the video...that was a metric for engagement”* (P21). Emotional accounts centered on interest and curiosity: *“My metric is mostly focused on if the things which are being said about it is of my interest. And when I hear about it, I want to know more about it”* (P22).

The interviews also identified influences on engagement that were distinct from participants’ definitions: audiovisual presentation and pacing, content novelty and jargon, prior familiarity, and structural clarity. These observations provide context for how participants formed their segment ratings.

7.2.2 Definition-Conditioned Calibration. Conditioning the modality scores on learners’ definition codes produces the strongest RQ2 performance (Table 5). The interaction model reaches 71.5% balanced 1-off accuracy and 1.124 macro-MAE, compared with 64.6% and 1.216 for score calibration. The fold-average improvement is 6.9 percentage points in balanced 1-off accuracy. The macro-MAE reduction is 0.092 in the fold averages.

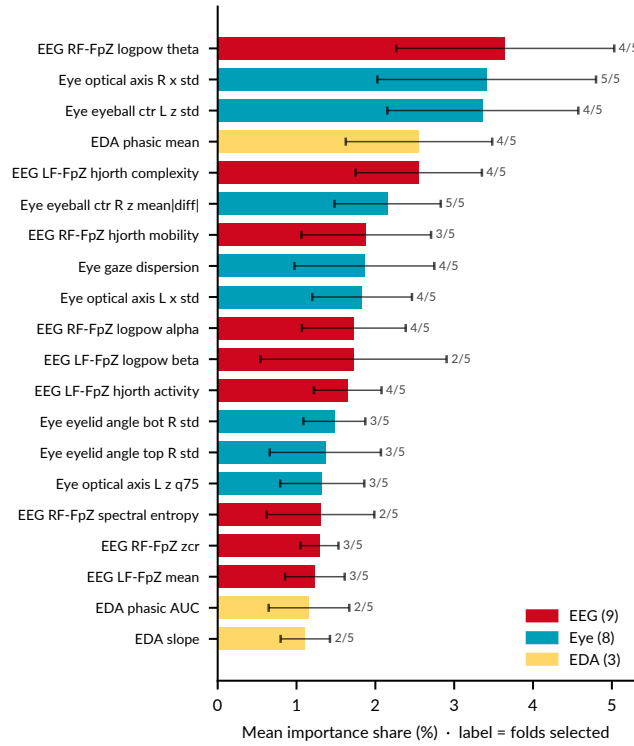


Fig. 5. **Feature importance underlying fixed-family AdaBoost selection.** For the AdaBoost top-20 sensitivity arm in Table 4, bars show the mean over five outer folds of each feature’s importance share in the initial 166-feature fit used for ranking; error bars show ± 1 SD across folds. Right-hand labels give the number of folds in which each feature entered the fold-local top 20. The 20 displayed features account for 38.6% of the mean total importance mass and comprise nine EEG, eight eye-tracking, and three EDA features. This figure corresponds to the fixed-family AdaBoost result ($75.9\% \pm 3.6$ balanced 1-off accuracy; 1.048 ± 0.120 macro-MAE), not a nested family-selection result.

Table 5. RQ2 ordinal prediction for eighteen coded participants under a six-fold participant-independent protocol. Values are equal-fold means. Score calibration is the primary comparator for the two definition-aware models; parentheses give the number of design columns. Adding definition–score interactions yields the strongest performance on both ordinal outcomes.

Model	Inputs	Balanced 1-off (%)	Macro-MAE
Constant code 3	No input (0)	60.0	1.400
Raw all-modality score	Score used directly (1)	63.2	1.216
Score calibration	Four scores + site (5)	64.6	1.216
+ Definition main effects	Above + three codes (8)	66.1	1.224
+ Definition–score interactions	Above + interactions (20)	71.5	1.124

Adding the definition codes as main effects gives 66.1% balanced 1-off accuracy and 1.224 macro-MAE. The larger gain from definition–score interactions is consistent with learners’ definitions modulating how the four upstream RQ2 scores map to self-reported ratings. With eighteen coded participants, this mechanism is an exploratory finding to test in a larger cohort.

8 Discussion

We summarize the main findings for our two research questions (Section 8.1), discuss their implications for engagement sensing and learning interfaces (Section 8.2), and outline the study’s limitations and directions for future work (Section 8.3).

8.1 Main Findings

RQ1: E³Sense provides participant-independent feasibility evidence for head-confined engagement sensing. The combined EEG, eye-tracking, and forehead EDA configuration predicted five-level engagement ratings from held-out learners after model development using data from both EDA sites. AdaBoost achieved the strongest ordinal point estimates: $75.0\% \pm 7.0$ balanced 1-off accuracy and 1.043 ± 0.158 macro-MAE (Table 3). These equal-fold means exceed the mode baseline by 12.0 percentage points in balanced 1-off accuracy and reduce macro-MAE by 0.277. The gains are descriptive and support the feasibility of the complete configuration under the tested protocol.

Model-family rankings depended on the prediction target. AdaBoost led the five-level ordinal evaluation, whereas LightGBM achieved the highest secondary binary macro-F1 at $58.9\% \pm 7.9$. AdaBoost also retained the strongest ordinal point estimates in the complete-case analysis, reaching $72.7\% \pm 4.8$ balanced 1-off accuracy and 1.066 ± 0.131 macro-MAE after recordings without measured EDA were excluded. Feature selection had mixed effects across model families (Table 4), supporting the complete 166-dimensional representation as the primary analysis.

RQ2: Learners’ definitions provide promising context for mapping sensor-derived scores to engagement ratings. Learners described engagement through understanding, sustained attention, interest, and emotion. These accounts span the cognitive, behavioral, and emotional dimensions of engagement and indicate that learners can draw on different criteria when assigning the same overall rating. In the exploratory analysis of eighteen participants, adding definition codes and their interactions with four frozen sensor-derived scores increased balanced 1-off accuracy from 64.6% to 71.5% and reduced macro-MAE from 1.216 to 1.124 relative to matched score calibration (Table 5). Adding definition main effects alone yielded 66.1% balanced 1-off accuracy and 1.224 macro-MAE. The stronger results with interactions are consistent with the hypothesis that definitions help condition how sensor-derived scores map to reported engagement. RQ2 uses its own cohort and evaluation protocol, its scores should be interpreted within that comparison rather than directly against RQ1.

8.2 Implications

8.2.1 Designing Engagement Sensing Around Head-Worn Devices. E³Sense provides a concrete sensing layout for combining EEG, eye tracking, and EDA within a head-worn configuration. Prior engagement systems such as SenseSeek and Effecti-Net combine EEG or eye tracking with EDA recorded at the wrist or fingers [21, 34], while Galea and HMDBioPad demonstrate co-located head-worn physiological instrumentation [6, 57]. E³Sense complements these systems by evaluating the complete head-confined combination against segment-level ratings from held-out learners.

This result motivates further integration of the three modalities into a shared form factor. Such integration could reduce the number of separately worn components and simplify fitting and synchronization. These are design opportunities to evaluate: the present configuration uses three devices, and its predictive performance does not establish comfort or ease of everyday use. The findings provide a basis for investigating whether a more integrated device can preserve useful engagement estimates while reducing wearing and setup demands.

8.2.2 Personalizing the Interpretation of Engagement. RQ2 connects engagement sensing with the multidimensional account of engagement in educational psychology [5, 23]. A single self-report can reflect several dimensions, and learners may emphasize them differently. This matters because physiological measurements acquire interpretive meaning through the labels with which they are associated [27, 28]. Learners’ descriptions make some of those rating criteria explicit and offer context for interpreting model outputs.

The resulting design implication is to include learners in defining what an engagement estimate represents. A future interface could briefly elicit whether a learner associates engagement with understanding, sustained attention, interest, or a combination of these, and use that information when mapping sensor-derived scores to ratings. The learner could review and update these criteria as tasks or goals change. Our interaction results motivate this approach, although prospective elicitation and its stability remain to be tested. The present codes were derived from interviews, they were not inferred as psychological traits from sensor data.

Definition context could also inform the design of learning support. For example, an interface could invite a learner who emphasizes understanding to review a difficult concept, or offer a pacing adjustment to someone concerned with sustaining attention. These are candidate interactions for future studies. The current models predict reported engagement and do not establish the cause of a low rating or which intervention would help.

8.2.3 Evaluating Performance Across Engagement Levels. Our findings emphasize the importance of evaluating less frequent engagement states. Ratings were concentrated in the engaged categories, making aggregate performance sensitive to the label distribution. The mode baseline illustrates this issue: always predicting code 3 receives within-one credit for codes 2–4 and attains 63.0% balanced 1-off accuracy in RQ1. A within-one score therefore requires interpretation alongside error magnitude and an appropriate sensor-free reference.

Macro-MAE averages absolute error equally across represented classes, preventing frequent ratings from dominating the measure [2]. Balanced 1-off accuracy applies the same class weighting to agreement within one rating level. We encourage engagement-sensing studies to report both measures alongside label distributions, per-class results, and constant baselines. For learning interfaces, model selection should also reflect the intended response: tracking graded changes and triggering a binary alert require different evaluations, as the change in model-family rankings between our ordinal and binary tasks illustrates.

8.3 Limitations and Future Work

8.3.1 Sample Size and Generalization. The study recruited thirty participants from one university community. RQ1 evaluated fifteen forehead participants, while RQ2 included eighteen participants who supplied explicit engagement definitions. Repeated segments increase the number of observations but do not increase the number of independent learners. The reported differences are descriptive, and these sample sizes limit claims about performance in broader populations. Participant-disjoint evaluation addresses transfer to unseen learners within this study, while the shared set of four educational videos leaves transfer to unseen materials untested. Larger studies should include more diverse learners and learning activities, with evaluation that holds out both participants and content.

8.3.2 Sensor Placement, Contribution, and Missing Data. EDA placement was confounded with collection batch: the first fifteen participants wore wrist EDA and the next fifteen wore forehead EDA. No participant wore both. Consequently, the study cannot isolate placement effects or quantify a performance trade-off between head-confined and distributed sensing. Moreover, every learned RQ1 model received the complete E³Sense representation, so its results do not

establish the contribution of each modality. A within-participant study recording both EDA sites, with counterbalanced conditions and prespecified sensor ablations, would address these questions.

Four participant-session recordings lacked usable EDA and required fold-local imputation in the primary analysis. The complete-case analysis retained AdaBoost’s ordinal ranking, but excluding those recordings changed both the training and test composition. It therefore cannot isolate the effect of imputation or establish robustness to sensor failure. Future evaluation should test predefined sensor-loss conditions and signal-quality variation.

8.3.3 Retrospective Ratings and Temporal Resolution. Participants rated segments retrospectively after each video. This preserved uninterrupted viewing, but ratings may reflect recall and impressions formed over the whole video. The model predicts these reports, which are not a direct measure of learning achievement. In addition, features summarize the first 120 seconds of segments lasting approximately 140–155 seconds. This partial coverage and temporal aggregation may obscure changes within a segment. Future work should compare labeling schedules and analysis-window lengths, assess agreement with complementary measures of engagement and learning, and test the temporal resolution needed for timely support.

8.3.4 Definition Elicitation and Model Complexity. RQ2 represents eighteen participant profiles using three overlapping definition indicators. The full calibration design contains twenty columns, including three definition main effects and twelve definition–score interactions. This complexity relative to the number of learners limits confidence in the stability of the observed gain. Holding upstream scores fixed makes the comparison specific to the added definition terms, but does not establish that the gain will reproduce or that definitions explain it independently of other participant or cohort differences.

Definitions were also elicited after the task and may reflect the completed study experience. Reducing interview accounts to three indicators loses variation within each category, and their stability across learning contexts was not assessed. A larger, preregistered study should elicit definitions before viewing, evaluate a smaller prespecified interaction set, and repeat elicitation across tasks. Any model selection should occur within participant-disjoint training folds, followed by evaluation on new learners and materials.

8.3.5 From Prediction to Everyday Learning Support. The evaluation concerns offline prediction during seated video viewing. It does not establish long-term wearability, performance during everyday movement, or the benefit of responding to predicted engagement. Future work should evaluate comfort, setup burden, signal reliability, and prediction latency in everyday learning settings. An interactive study could then compare sensing-informed support with a suitable control, measuring learning outcomes and interruption burden as well as prediction quality. This would test whether the sensing and definition context investigated here translate into useful support for learners.

9 Conclusion and Future Work

E³Sense co-locates EEG, eye tracking, and forehead EDA as a head-confined sensing configuration. AdaBoost using the complete E³Sense representation achieves $75.0\% \pm 7.0$ balanced 1-off accuracy and 1.043 ± 0.158 macro-MAE on held-out forehead participants. Its balanced 1-off fold mean is 12.0 percentage points above mode. These descriptive results provide preliminary feasibility evidence for the complete head-confined configuration under the tested protocol. Because every learned RQ1 model uses the full E³Sense representation, this comparison does not estimate the contribution of any individual sensor. Controlled ablations in a larger cohort remain future work.

Learners' exit-interview accounts are an initial, exploratory check on what that sensing result predicts. In a coded-subsample analysis, the interaction arm over frozen, cross-fitted upstream scores attains 71.5% balanced 1-off accuracy versus 64.6% for score calibration, with macro-MAE decreasing from 1.216 to 1.124. This result is consistent with the construct E³Sense's models predict being something learners can name, and naming it sharpening the prediction (see §8).

Future work should test sensor attribution, transfer to new learning contexts, and definition-conditioned calibration in larger cohorts (§8). Taken together, these results indicate that a fully head-confined sensor set can predict self-reported engagement, with learners' own accounts of what engagement means to them providing a source of corroborating context sensing design has not yet drawn on.

Data and Code Availability

Analysis code and de-identified derived features supporting the reported results will be released upon request.

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